



Washington University in St. Louis

JAMES MCKELVEY SCHOOL OF ENGINEERING

SP2024.E37.MEMS.412.01: Design of Thermal Systems

Final Project Report: Ice Rink Cooling System

Course Instructor: Patricia Weisensee

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Group 14: FEAR (8:30-9:50am)

We hereby certify that the lab report herein is our original academic work, completed in accordance with the McKelvey School of Engineering and Undergraduate Student academic integrity policies, and submitted to fulfill the requirements of this assignment:

Freddy Phillips

Eleni Kambouris

Aidan Tierney

Rachel Foulk

ABSTRACT: *The primary objective of this lab report was to conduct an exploratory analysis of the thermal systems present in an Olympic sized ice rink and optimize the parameters in pursuit of the best coefficient of performance while varying the mass flow rate of coolant fluid. The working fluids selected for this task of optimization were ammonia NH_3 as the primary working fluid and carbon dioxide CO_2 as the secondary working fluid. FEAR discovered that by varying the mass flow rate of CO_2 between 2 L/min and 100 L/min increased the heat exchanged between the two refrigeration cycles. The highest rate of heat transfer was found to be 0.7561 kW at a mass flow rate of 0.0030 kg/s, necessitating a chiller tubing length of 50.6098 m. Surprisingly, no strong correlation appeared between varying the the mass flow rate of the coolant fluid and the system's coefficient of performance, which stayed constant at 6.6896. This indicates that other cycle parameters have more significant influence over the coefficient of performance, which will require further exploratory analysis to determine a procedure to optimize. Whether or not the mass flow rate of the cooling fluid needs to be increased depends on the desired heat transfer as well as pipe and cycle component limitations. These findings provide a foundation for future research aimed at refining the optimization of ice rink cooling systems, ultimately contributing to the development of more efficient and sustainable thermal solutions.*

1 INTRODUCTION

Ice rinks, a staple cultural center in many communities, rely on sophisticated cooling systems to maintain their frozen surfaces. There are a variety of approaches to complete this goal, however indirect refrigeration systems stand out as the most common approach. Unlike direct systems where the refrigerant directly interacts with the cooled surface, indirect systems utilize a secondary liquid, often carbon dioxide, to facilitate heat exchange [1]. In this method, a liquid refrigerant, commonly ammonia, absorbs heat from the carbon dioxide. Carbon dioxide, with its ability to remain in a liquid state at low temperatures and moderate pressures, plays a pivotal role in indirect refrigeration systems [2]. As the carbon dioxide flows through the system's piping network, it absorbs heat from the ice rink surface, effectively drawing heat away and maintaining the desired frozen state. This indirect nature ensures that the refrigerant does not directly contact the surface being cooled, thus increasing efficiency.

The heart of the indirect refrigeration system comprises three key components: the chiller, compressor, and condenser [1]. In the chiller, the liquid refrigerant, ammonia in this case, absorbs heat from the carbon dioxide returning from the ice rink surface, causing it to boil and vaporize. This vaporized refrigerant is then drawn into the compressor, where it undergoes pressurization, raising its temperature significantly [1]. As the compressed refrigerant enters the condenser, it begins to condense, releasing heat and returning to a liquid state. This cooled liquid refrigerant is then circulated back to the chiller, ready to absorb heat once again, thus completing the refrigeration cycle.

Although the refrigeration cycle has now ended in theory, the most applicable piece of this technology is yet to come. While the refrigeration cycle operates in the primary refrigeration site, the recently chilled carbon dioxide is pumped into pipes embedded within the rink floor. These copper pipes, inter-laid inside the entire surface area of the rink, facilitate the transfer of heat from the ice surface to the carbon dioxide. Due to heat always flowing from hot to cold, the colder carbon dioxide wicks away heat from the ice's surface, maintaining it at the desired temperature of approximately -4°C. Subsequently, the now warmer carbon dioxide returns to the chiller, where it releases the absorbed heat back to the refrigerant before repeating the cycle.

These interactions sum into a complete cooling system for an ice rink. The primary objective of this type of system is to maximize efficiency while attempting to minimize energy consumption and operational costs. In this report, group FEAR will analyze and attempt to optimize such a system, aiming to enhance its performance and reliability while meeting operational constraints and considering real world limitations.

2 Literature Review

The examined research offers valuable insights into the design and operation of ice rink cooling systems, particularly those employing ammonia as the primary working fluid and carbon dioxide (CO₂) as the secondary working fluid.

The literature emphasizes the prevalence of indirect refrigeration systems in ice rink applications, wherein a heat transfer fluid, often ammonia, is pumped into the rink pipes from a storage tank [1]. CO₂ has been increasingly utilized as a secondary fluid in such systems [1], offering advantages over traditional brine solutions. Unlike brine, CO₂ is a phase-changing fluid, therefore

lowering the pumping power required. Additionally, CO₂'s high operating pressure allows for the use of metal pipes, which are more robust compared to plastic pipes commonly used with brine [2].

The two major contributions to heat loads present in indoor ice rinks are conductive and convective. Conductive loads from the ground and piping are the two of the most significant factors affecting system design and operation. Proper insulation and design to minimize heat gain are essential for maintaining ice quality and minimizing energy consumption [2]. The literature also discusses the importance of circulation rate of CO₂ to facilitate efficient heat transfer [1]. Convective loads from air to the ice surface represent a significant portion of the total heat load, potentially exceeding 28 [2]. Factors such as air temperature, velocity, and relative humidity significantly influence convective heat transfer [3]. Proper design considerations, such as minimizing air movement across the ice surface and banning smoking in the arena to reduce ventilation [2], are crucial for minimizing convective loads and optimizing system performance.

Furthermore, research findings suggest that the choice of piping material and diameter, as well as the optimization of tube pitch, are crucial to optimize system performance. Copper tubes with smaller diameters and reduced pitch have been shown to offer better heat transfer performance compared to larger steel pipes [2]. However, practical considerations such as cost and installation constraints must be weighed when selecting optimal system parameters [4].

Overall, the literature emphasizes the complexity and importance of various factors in the design and operation of ice rink cooling systems, with a focus on maximizing thermodynamic efficiency, minimizing energy consumption, and minimizing economic costs. These insights will inform FEAR's analysis and optimization efforts in this research project.

3 Assumptions/Boundary Conditions

- System operates under an indirect refrigeration system. In this system there are two fluids acting in parallel. The refrigerant fluid will be ammonia and the secondary fluid will be carbon dioxide (CO₂) coupled by one main heat exchanger.
- The primary cycle operates as a reverse rankine cycle. The secondary cycle operates as a basic heat transfer cycle [3].

- State assumptions are as follows: State 1 (saturated vapor), State 2 (superheated vapor), State 3 (saturated liquid), State 4(two-phase), State 5 (saturated Liquid), State 6 (saturated Liquid), State 7 (saturated vapor), State 8 (saturated vapor) [5].
- The following temperatures will be assumed in according to relevant research for the primary refrigeration cycle (Ammonia): $T_1 = -11^{\circ}\text{C}$, $T_3 = 38^{\circ}\text{C}$. [5].
- The following temperatures will be assumed in according to relevant research for the secondary refrigeration cycle (CO2): $T_5 = -10^{\circ}\text{C}$, $T_6 = -10^{\circ}\text{C}$, $T_7 = -10^{\circ}\text{C}$, $T_8 = -10^{\circ}\text{C}$. The ice is held at -4°C . The CO2 in the ice rink loop is undergoing phase change [2] , [5].
- The pressure at state 2 is defined as $P_2 = -1554.9 \text{ kPa}$ [1].
- In the chiller, the CO2 and NH3 undergo phase change [5]. The chiller operates as a concentric tube heat exchanger (thin-walled tubes) in counterflow with ammonia flowing in the annulus and CO2 flowing in the inner diameter. The inner and outer diameters are defined as $d_i = 0.6 \text{ in}$ and $d_o = 1 \text{ in}$. The flow of the CO2 and NH3 in the chiller is assumed to be developed with constant flux.
- There is a pressure drop due to major loss in the ice rink. The specifications of the ice rink and piping system are as follows: Olympic ice rink dimensions: 200ft x 100ft [6], Pipe spacing: 4 inches [7], Pipe diameter: 1 in [7], Tube Passes: 2 [8], Pipe Material: Copper [2].

Schematics and Diagrams

3.1 Diagrams Describing Ice Rink Cooling System. As a preliminary step to the analysis of the refrigeration system, a schematic was created of the complete cycle. The states in this diagram were labeled somewhat arbitrarily, but attempted to remain consistent with other current research on the same thermodynamic cycles. For ease of discussion and examination, the complete system was divided into two subsections: system 1 which uses ammonia (NH_3) as it's working fluid and system 2 which utilizes carbon dioxide (CO_2) as it's working fluid. These two sections interact through a heat exchanger which will be referred to as the chiller.

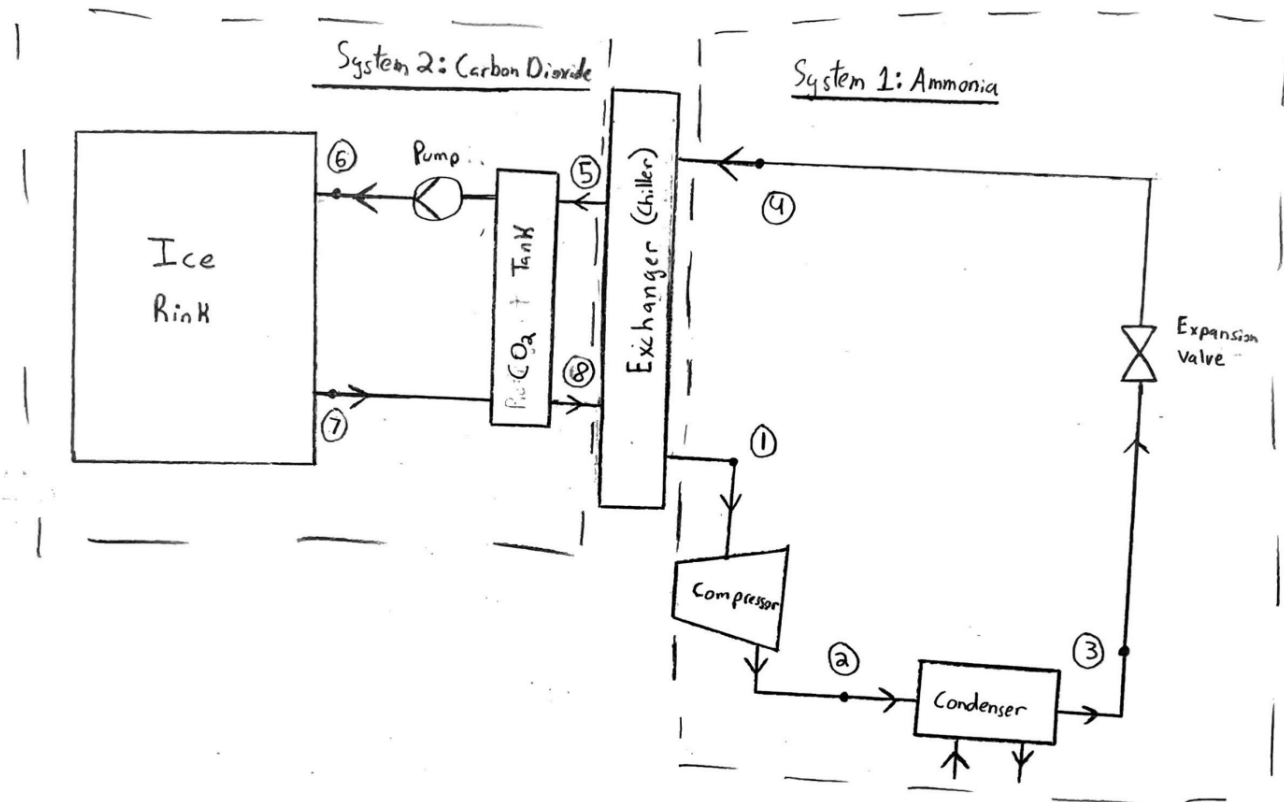


Figure 1 Complete Schematic of Refrigerant Cycle.

As declared previously, state's 1 through 4 express ammonia as a cold gas, hot pressurized gas, cold pressurized gas, and cold liquid respectively. Similarly, carbon dioxide is described with state's 5 through 8, however in contrast to ammonia there is no phase change in across system 2 only a relatively small temperature gradient. These states are coldest liquid, second coldest pressurized liquid, relatively warm liquid, and warmest liquid.

1 → 2:	adiabatic compression	$T \uparrow, P \uparrow$
2 → 3:	isobaric cooling	$T \downarrow, P -$
3 → 4:	isenthalpic expansion	$T \downarrow, P \downarrow$
4 → 1:	isobaric heating	$T \uparrow, P -$

Figure 2 Handwritten Table Displaying Refrigerant Cycle's Steps and the Effect on Temperature and Pressure.

For clarity, the table above was constructed showcasing the cause and effect of the refrigeration

cycle's effect on ammonia's state. A similar table was not constructed for carbon dioxide due to its simplicity.

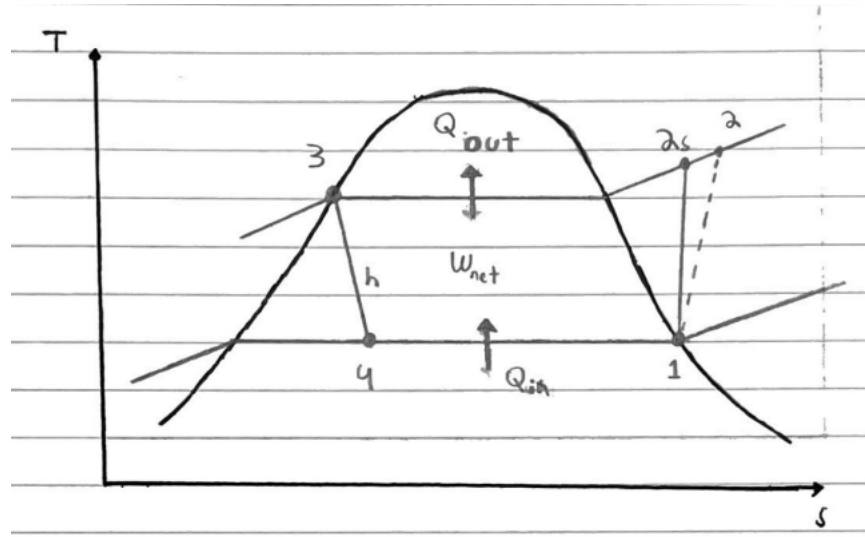


Figure 3 T-s Diagram of Refrigeration Cycle.

This diagram was modeled after a similar one present in [3]. As can be seen in the diagram, despite the difference in temperature and pressure in states 3 and 4, enthalpy remains constant 2. Under normal circumstances, T_2 would need to be determined by calculation but [2] provided a rigorous proof for this temperature. Q_{out} is the heat exiting the system through the condenser, which is more commonly referred to as Q_L . Q_{in} is the heat entering system 1 via the heat exchanger (chiller) as it leaves the carbon dioxide. The next step in the analysis is to consider what these relations look like on a T-x diagram, which represents the change in temperature over the distance of the entire chiller. The next diagram displays a stylistic diagram of the movement of ammonia and carbon dioxide through the heat exchanger.

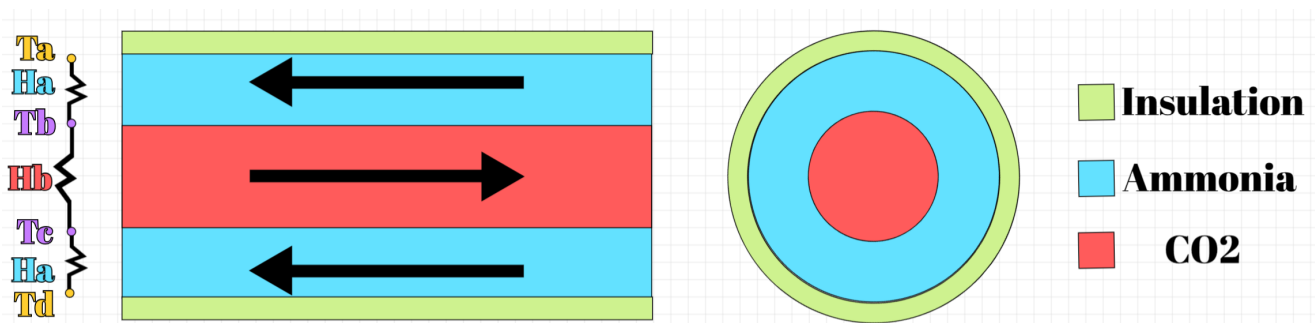


Figure 4 Counter Flow Concentric Tube Heat Exchanger Diagram.

The diagram above depicts the inner workings of a section of tube in the primary heat exchanger in the refrigerant system. The arrows denote the mass flow rate in the localized area with the colors representing their respective working fluid or pipe casing. To the left of the diagram is a heat transfer resistance network. This type of heat exchanger and configuration were chosen based on a prior lab conducted by FEAR for the sake of thermal efficiency and familiarity.

4 Equations

$$\begin{array}{lll}
 \textcircled{1} \quad \beta = \frac{q_L}{w_c} & \textcircled{2} \quad Re_{CO_2} = \frac{q(\dot{m}_{CO_2}/N)}{\pi d_i \mu_{CO_2}} & \textcircled{3} \quad f = \frac{64}{Re_{CO_2}} \\
 \textcircled{4} \quad A = \frac{\pi d^2}{4} & \textcircled{5} \quad V = \frac{\dot{m}_{CO_2}}{\rho A} & \textcircled{6} \quad P_{\text{major loss}} = f \left(\frac{L}{D} \right) \left(\frac{\rho V^2}{2} \right) \\
 \textcircled{7} \quad W_{\text{pump}} = V \left(\frac{P_{\text{major loss}}}{1000} \right) & \textcircled{8} \quad Q_{\text{ice sink}} = \dot{m}_{CO_2} (\Delta h) & \textcircled{9} \quad W_{\text{comp}} = \dot{m}_{NH_3} h_2 - \dot{m}_{NH_3} h_1 \\
 \textcircled{10} \quad Q_{NH_3 \text{ chiller}} = (\dot{m}_{NH_3}) (h_1 - h_4) & \textcircled{11} \quad Q_{\text{chiller}} = \dot{m}_{CO_2} (\Delta h) & \textcircled{12} \quad \dot{m}_{NH_3} = \frac{Q_{\text{chiller}}}{Q_{NH_3 \text{ chiller}}} \\
 \textcircled{13} \quad h_2 = h_1 - \eta_{\text{isem}} (h_1 - h_{2s}) & \textcircled{14} \quad Q_{\text{cond}} = \dot{m}_{NH_3} (h_2 - h_3) & \textcircled{15} \quad D_h = (d_o - d_i) \\
 \textcircled{16} \quad Re = \frac{\rho u_m D_h}{\mu} & \textcircled{17} \quad u_m = \frac{\dot{m}}{\rho A} & \textcircled{18} \quad Re = \frac{D_h \dot{m}}{\mu N \left(\frac{\pi (d_o^2 - d_i^2)}{4} \right)} \\
 \textcircled{19} \quad \frac{1}{U} = \frac{1}{h_{NH_3}} + \frac{1}{h_{CO_2}} & \textcircled{20} \quad h_{CO_2} = \frac{N_{CO_2} k_{CO_2}}{d_i} & \textcircled{21} \quad h_{NH_3} = \frac{N_{NH_3} k_{NH_3}}{d_h} \\
 \textcircled{22} \quad Q = UA \Delta T_{lm} & \textcircled{23} \quad \Delta T_{lm} = \frac{\Delta T_2 - \Delta T_1}{\ln \left(\frac{\Delta T_2}{\Delta T_1} \right)} & \textcircled{24} \quad \Delta T_2 = T_{h,i} - T_{c,o} \\
 & & \Delta T_1 = T_{h,o} - T_{c,i} \\
 \textcircled{25} \quad A = N \pi D L & \textcircled{26} \quad C_{NH_3} = \dot{m}_{NH_3} c_{p,NH_3} & \textcircled{27} \quad C_r = \frac{C_{min}}{C_{max}} \\
 \textcircled{28} \quad L_{\text{chiller}} = \frac{Q}{N \pi d_h U \Delta T_{lm}} & \textcircled{29} \quad C_{CO_2} = \dot{m}_{CO_2} c_{p,CO_2} & \\
 \textcircled{30} \quad N = \frac{(D \cdot \text{width} + 2(\text{spacing}))}{D + \text{spacing}} & &
 \end{array}$$

Figure 5 List of Equations That Will be Used to Find Design Parameters.

5 RESULTS & DISCUSSION

5.1 Plots for Optimization. For this design, the mass flow rate of the CO₂ was chosen as the input variable, and the coefficient of performance of the primary refrigeration cycle and the rate of heat transfer in the ice rink was chosen to be optimized. Fig. 6, shows the plot used to optimize the rate of heat transfer in the ice rink as a function of the CO₂ mass flow rate.

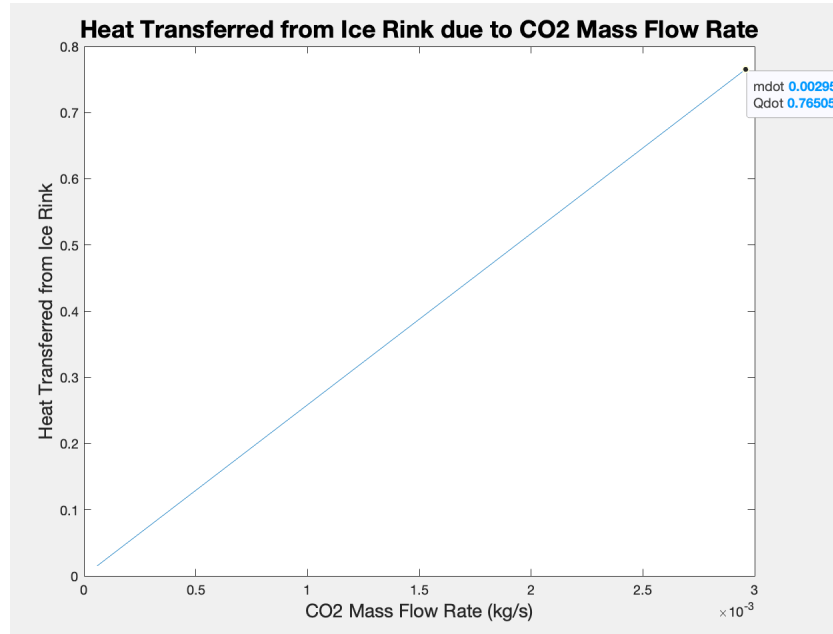


Figure 6 Rate of Heat Transfer in Ice Rink Optimization Plot.

As shown by Fig. 6, the rate of heat transfer in the ice rink appears to have a positive linear relationship with the CO₂ mass flow rate. As the CO₂ mass flow rate increases, so does the heat transferred from the ice rink. The maximum value of heat transfer in the ice rink is 0.7651 kW and occurs when the CO₂ mass flow rate has a value of 0.0030 kg/s, optimizing the heat transfer.

Additionally, Fig. 7 shows the plot used to optimize the coefficient of performance of the primary refrigeration cycle as a function of the CO₂ mass flow rate.

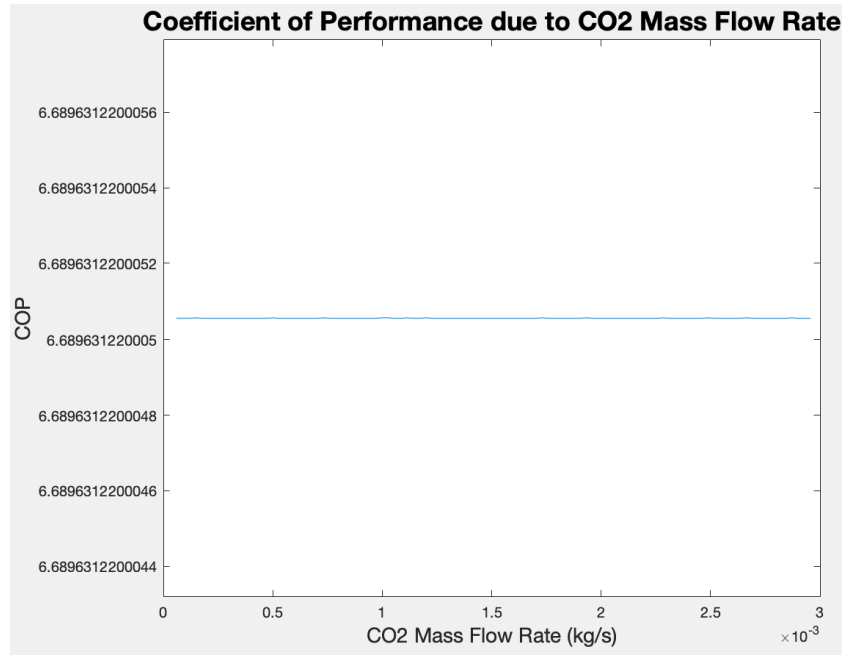


Figure 7 Coefficient of Performance Optimization Plot.

As shown by Fig. 7, the coefficient of performance (COP) appears to have a relatively constant trend with the CO2 mass flow rate. As the CO2 mass flow rate increases, the COP largely stays constant. Therefore, for the values of CO2 mass flow rate chosen, the COP has a value of 6.69.

Since the COP remains constant, it is better to optimize the rate of heat transfer in the ice rink. Additionally, optimizing the cooling load of the ice rink will ensure the best ice quality and conditions. Because of this, the optimal mass flow rate is 0.0030 kg/s.

5.2 Final Answers. Table 1 details the final answers after optimization.

Table 1 Final answers.

CO2 Mass Flow Rate	0.0030 kg/s
NH3 Mass Flow Rate	0.00072 kg/s
Heat Transfer Rate: Ice Rink	0.7651 kW
Heat Transfer Rate: Chiller	0.7651 kW
Length of Tubes in Chiller	50.6098 m
Coefficient of Performance (Primary Cycle)	6.6896
Number of Tubes in Ice Rink	241 tubes

5.3 Sanity Check. The final answers were checked both theoretically using sources and practically using logic to make sure they made sense. First, the heat transfer rate of the ice rink is the same as the heat transfer rate of the chiller, 0.7651 kW. This makes sense because the CO₂ is undergoing the same enthalpy change in the ice rink as it is in the chiller while the mass flow rate remains constant. Therefore, it is expected for these values to be the same. Regarding the mass flow rate of the CO₂, it makes sense that a higher mass flow rate would optimize the heat transfer in the ice rink. As shown in Fig. 5, the equation for the heat transfer in the ice rink is directly proportional to the CO₂ mass flow rate. Therefore by increasing the CO₂ mass flow rate, the heat transfer should and does increase. Additionally, a research paper on carbon dioxide in ice rink refrigeration found the coefficient of performance to be 4.9 for an NH₃/Brine system [5]. Comparatively, the coefficient of performance was 6.6896 for our NH₃/CO₂ system. Because brine is known to be less effective at transferring heat than CO₂, it makes sense that our system would have a higher coefficient of performance. The length of the tubes calculated in our chiller was 50.6098 m. Comparing this to industrial chillers, such as 200 ton chiller typically used for most Olympic ice rinks, the total length of the chiller is 7.62 meters, meaning the length calculated for our chiller is a little large [9]. This can be due to a multitude of factors such as a difference in components in both the primary and secondary refrigeration cycle and broad simplifications of the system. Additionally, most chillers, such as the one being compared to, are shell and tube heat exchangers while ours is a concentric tube heat exchanger. This could account for the difference in length.

5.4 Financial and Environmental Impact Analysis. The financial and environmental impacts of any engineering project are critical considerations in the design and operation processes. Naturally then, economical optimization is a key aspect of ice rink system design. The two emergent considerations that dominate cost reduction of operational costs lie in ventilation and pipe optimization. As could be expected, ventilation constitutes one of the largest factors in the annual operational and maintains costs for an ice rink. The analysis of copper tubes in ice rink applications aims to reduce initial investment and keep operational costs low. The optimization process involves determining the optimum pitch of the copper tubes, which balances investment costs with operational efficiency.

Maintaining good water quality through water treatment not only ensures the longevity of the

system but also enables lower flood water temperatures and reduces the required volume of water, decreasing cost and environmental impact [4]. Additionally, implementing measures such as banning smoking in the arena helps minimize ventilation needs, thereby reducing energy consumption and operational costs associated with humidity control [2].

The research highlights the trade-off between initial investment and operational costs associated with varying the pitch of the copper tubes [2]. Reducing the pitch raises the initial cost of installment but lowers the evaporation temperature, resulting in reduced operational costs [4].

Overall, considering both financial and environmental impacts is essential while designing sustainable and economically viable ice rink cooling systems. Optimizing system parameters and implementing efficient practices, significant cost savings can be achieved while minimizing environmental footprint.

6 Conclusion

This report investigates the effects of varying the mass flow rate of a secondary refrigeration cycle's cause and effect relationship on the rate of heat transfer and the coefficient of performance of the cycle. These outputs were then optimized for use in an ice rink cooling system that is coupled by a heat exchanger with NH_3 as the primary and CO_2 as the secondary working fluid. The mass flow rate of the CO_2 was varied between 2 L/min and 100 L/min, resulting in a positive trend between heat transferred and mass flow rate with the most heat being transferred at the highest mass flow rate. Conversely, mass flow rate does not seem to have a very large effect, if any, on the coefficient of performance. Ultimately, FEAR determined the optimal CO_2 mass flow rate to be 0.003 kg/s. This facilitated a heat transfer rate between the systems of 0.7651 kW, requiring a chiller tubing length of 50.6098 m, producing a COP of 6.69. An intriguing and promising way to expand the scope of this analysis could come in the form of exploring a wider range of secondary working fluids, such as brine or creating a similar working fluid in pursuit of finding a perfectly optimized fluid for a comparable cooling cycle promising advancements in efficiency and performance. In summation, this report determined a positive relation between mass flow rate and heat transfer rate while determining a minimal, if any, effect of mass flow rate on the coefficient of performance in a refrigeration cycle.

References

- [1] Stubbings, C., 2023, “Ice rinks: Refrigeration on a big scale. what’s under the surface you skate on?: Chart Industries,” <https://www.chartindustries.com/Articles/Ice-Rinks-Refrigeration-On-A-Big-Scale>
- [2] Shahzam, K., 2006, “Ice rink co2 + copper tube design - ice skating resources,” https://www.yumpu.com/en/document/read/33369040/ice-rink-co2-copper-tube-design-ice-skating-resources#google_vignette
- [3] Borgnakke, C. and Sonntag, R. E., 2014, *Fundamentals of thermodynamics*, 10th ed., Wiley.
- [4] Sutton, J., 2023, “Thermodynamics of Ice Arena Geometry, Low- Emissivity of Building Materials, and Natural Ventilation: An Energy Efficiency Study of an Ice Arena in Rochester, NY,” <https://repository.rit.edu/cgi/viewcontent.cgi?article=10791&context=theses>
- [5] Nguyen, T., 2013, “Carbon dioxide in ice rink refrigeration,” <https://kth.diva-portal.org/smash/record.jsf?pid=diva2%3A604747&dsid=6813>
- [6] Condor, B., 2021, “Measuring and marking up,” <https://www.nhl.com/kraken/news/dimensions-and-markings-of-regulation-ice-sheets-322765100>
- [7] 2022, “Ice Rinks - Ammonia Standard System,” https://www.arcat.com/sdspeccs/htm04/13_18_00cim.htm
- [8] 2020, <https://www.youtube.com/watch?v=M-twpk6ivjc>
- [9] 2022, “Carolina Cat Power Generation 2022,” <https://carolinacat.com/power-generation/rental-equipment/power-systems/chillers/200-ton-chiller/>
- [10] Gerhart, A. L., Gerhart, P. M., and Hochstein, J. I., 2021, “Munson, young, and Okiishi’s fundamentals of Fluid Mechanics,” .
- [11] Cavallo, C., 2024, “All About Double Pipe Heat Exchangers – What You Need To Know,” .

A MATLAB Code

```
1 %% ICE Rink Cooling System
2
3 %% Loop!
4
5 %% Set Up Knowns:
6 T5 = -10; % C
7 T6 = -10;
8 H67 = 258.61; %kJ/kg
9 T7 = -10;
10 T8 = -10;
11
12 T1 = -11; % C
13 T2 = 100; % C (chart industries)
14 T3 = 38; % C
15 h1 = 1430.8; %kJ/kg
16 h3 = 371.43; %kJ/kg
17 h4 = h3; %kJ/kg (expansion valve)
18 s1 = 5.4673; %kJ/kg*K
19
20 rho_carbon = 1.775; % kg/m^3
21 v_carbon = 0.01303; % m^3/kg
22 cp_Carbon = 0.842; % kJ/kg-K
23 cp_Ammonia = 2.130; % kJ/kg-K
24
25 eqroughness = 0.0015; %mm
26 D = 0.0254; %m
27 L = 60.96; %m
28 Spacing = 0.1016; %m
29 N_ice = (30.48 + 2*Spacing) / (D + Spacing);
30
31 mu_Co = 140*10^(-7); %Ns/m^2
32 Pr_Co = 0.765;
33 k_Co = 7.36*10^(-6); %m^2/s
34
35 mu_Am = 101.5*10^(-7); %Ns/m^2
```

```

36 Pr_Am = 0.887;
37 k_Am = 14.7*10^(-6); %m^2/s
38
39
40 %Setting up Loop Variables:
41 Lmin_conversion = (1/60)*(1/1000)*rho_carbon;
42 mCstart = 2*Lmin_conversion; % kg/s
43 mCend = 100*Lmin_conversion; % kg/s
44 mC = linspace(mCstart,mCend,100);
45
46 efficiency_CO2 = zeros(length(mC),1);
47 efficiency_ammonia = zeros(length(mC),1);
48 QIceRink = zeros(length(mC),1);
49 WIcePump = zeros(length(mC),1);
50
51 %%
52 for k = 1:length(mC)
53
54 % Ice Rink CO2 Cycle
55     %% Pump
56     Re_CO2cycle = (4*(mC(k)/N_ice))/(pi*D*mu_Co);
57     frictionfactor = 64/Re_CO2cycle;
58
59     A = (pi/4)*(D)^2; %m^2
60     VCO2 = (mC(k)/N_ice)/(rho_carbon*A); %m/s
61     pmajorloss = frictionfactor*(L/D)*((rho_carbon*(VCO2^2))/2);
62
63     WIcePump(k) = v_carbon*(pmajorloss/1000); %kW
64
65 %Heat Exchange
66     QIceRink(k) = mC(k)*(H67); %kW
67     QHeatExchanger(k) = mC(k)*(H67); %kW
68
69 %Efficiency
70     efficiency_CO2(k) = QHeatExchanger(k)/(WIcePump(k));
71
72 %% Ammonia Rankine Cycle
73     %Evaporator

```

```

74     % mh4 + Ql = mh1
75     qheatExchanger(k) = (h1-h4); %kJ
76     mAmmonia(k) = QHeatExchanger(k)/qheatExchanger(k); %kg/s
77
78     %Compressor
79     %mh1 + Wc = mh2
80     nisentropic = 0.74;
81     s2s = s1;          %kJ/kg*K (@100 C and 1554.9/1600kPa)
82     h2s = 1644.8;      %kJ/kg
83
84     h2 = h1 - (nisentropic*(h1-h2s)); %kJ/kg
85     WAmmoniaCompressor(k) = mAmmonia(k)*(h2-h1); %kW
86
87     %Condenser
88     %mh2 = mh3 + qH
89     QAmmoniaCondenser(k) = mAmmonia(k)*(h2-h3);
90
91     %COP
92     efficiency_ammonia(k) = QHeatExchanger(k)/(WAmmoniaCompressor(k));
93
94     % Heat Exchanger
95
96     % Constants
97     do = 0.0254;
98     di = 0.01524;
99     N = 200;
100    nh = 0.4;
101    nc = 0.3;
102
103    % CO2 (inner tube)
104
105    % Reynolds Number
106    Re_Co = (4*mC(k))/(N*pi*mu_Co*di);
107
108    % Nusselt Number
109    Nu_Co = 4.36;
110
111    % Heat Transfer Coefficient

```

```

112     h_Co(k) = (Nu_Co*k_Co)/di;
113
114 % Ammonia (outer tube)
115 % Reynolds Number
116 dh = do-di;
117 Re_Am = (dh*mAmmonia(k))/(mu_Am*N*(pi/4)*(do^2-di^2));
118
119 % Nusselt Number
120 Nu_Am = 4.36;
121
122 % Heat Transfer Coefficient
123 h_Am(k) = (Nu_Am*k_Am)/dh;
124
125 % Length of HE
126 % Using Log-Mean (not required since temp difference is same on both
127 % sides)
128 % T4 = T1;
129 % delT1 = T5-T4;
130 % delT2 = T8-T1;
131 % Tlm_s = (delT2-delT1)/log(delT2/delT1);
132 Tlm_s = 1;
133
134 %Overall HT coefficient
135 U(k) = ((1/h_Am(k))+(1/h_Co(k)))^-1;
136
137 %Length of HE
138 L_HE(k) = QHeatExchanger(k)/(N*pi*di*U(k)*Tlm_s);
139
140 end
141
142 %%
143 %Optimization Graphs
144 figure(1);
145 plot(mC,efficiency_ammonia)
146 xlabel('CO2 Mass Flow Rate (kg/s)')
147 ylabel('COP')
148 title('Coefficient of Performance due to CO2 Mass Flow Rate')
149 hold on

```

```

150
151 figure(2);
152     plot(mC,QIceRink)
153     xlabel('CO2 Mass Flow Rate (kg/s)')
154     ylabel('Heat Transferred from Ice Rink')
155     title('Heat Transferred from Ice Rink due to CO2 Mass Flow Rate')
156     hold on
157
158 % figure(3);
159 %     plot(mC,WIcePump)
160 %     xlabel('CO2 Mass Flow Rate (kg/s)')
161 %     ylabel('Work of CO2 Pump')
162 %     title('Work of CO2 Pump due to CO2 Mass Flow Rate')
163 %     hold on

```